



Coventry University Group

Scarborough

Welcome to CU Scarborough Risk Management

Risks

- Understanding the likely risks any project might face, having a plan in place to deal with the ones you can, and perhaps can't anticipate, are all part of the key project management skillset.
- Example: a simple risk exercise can save hours of 'recovery' time in a project, by anticipating what might go wrong and what might be done to rectify this. A recent project assumed that all users of a system could be trained together in one place, which at the time was reasonable, but the risk turned issue was a cut in travel budget. If virtual training had initially been considered as a plan B, this may have been a better solution

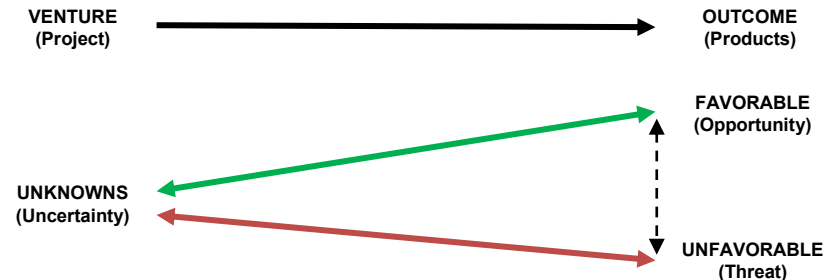
Introduction

- Project Risk Management is frequently overlooked, yet it is one of the more critical elements to successful project delivery. Generally, delivering a project's defined scope on time and within budget are characteristics of project success (KPMG, 2014).
- Unfortunately, these success factors are often not achieved, especially for large complex projects where both external influences and internal project requirements may change significantly over time.
- Project Risk Management is a continuous process of identifying, analysing, prioritising and mitigating risks that threaten a project's likelihood of success in terms of cost, schedule, quality, safety and technical performance.
- Organisations and owners often consider project risk management activities as “nice to have” on a project rather than as a core component of project controls.
- Additionally there is some confusion between organisations and project teams as to what exactly constitutes risk management activities.

Risk Management Overview: What is Risk?

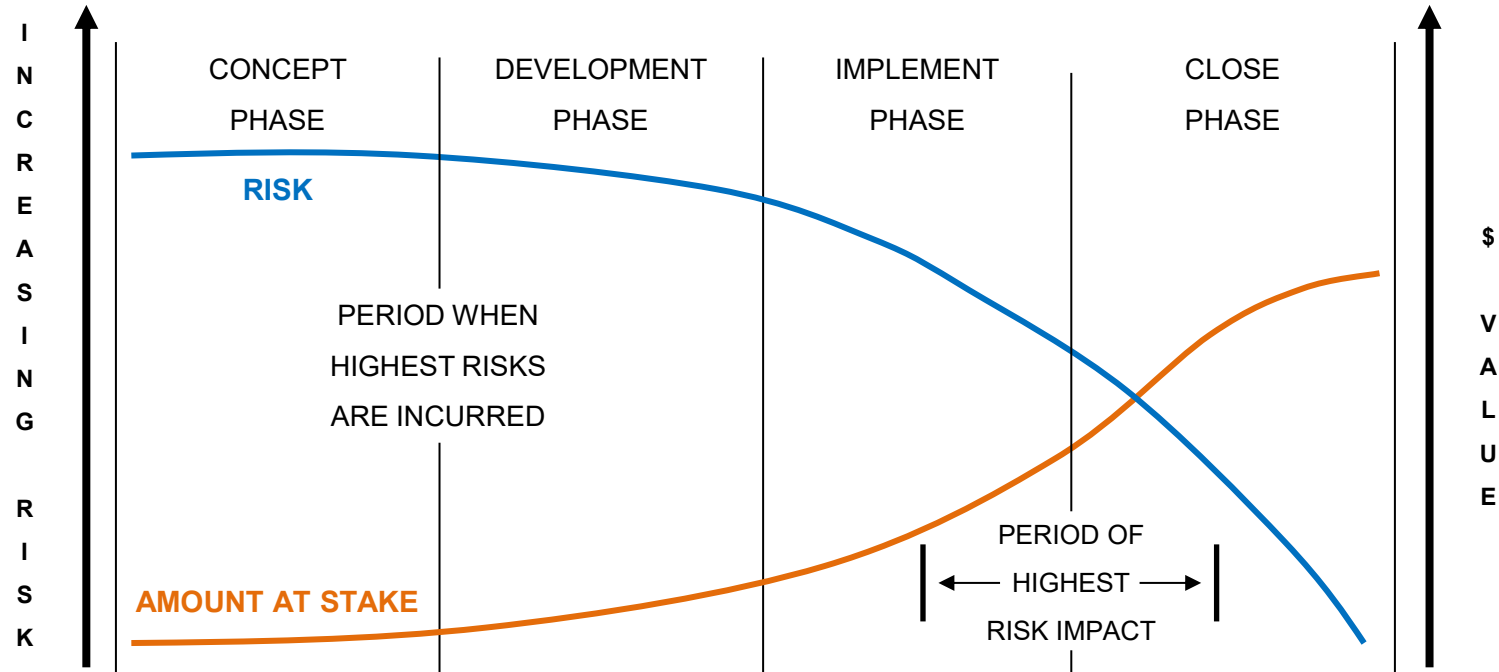
- The set of hindrances that threaten the pursuit of business objectives (PMI BoK, 2004).
- Uncertainty inherent in plans and the possibility of something happening (i.e. a contingency) that can affect the prospects of achieving business or project goals (Gaudenzi and Borghesi, 2013).
- Risk taking in projects is inevitable since projects are enablers of change; and change introduces uncertainty, hence risk. It consists of a combination of the probability of a perceived threat or opportunity occurring, and the magnitude of its impact on objectives, where:

- **Threat** is used to describe an uncertain event that could have a negative impact on objectives.
- **Opportunity** is used to describe an uncertain event that could have a favourable impact on objectives.



Project Lifecycle

Risk vs. Amount at Stake

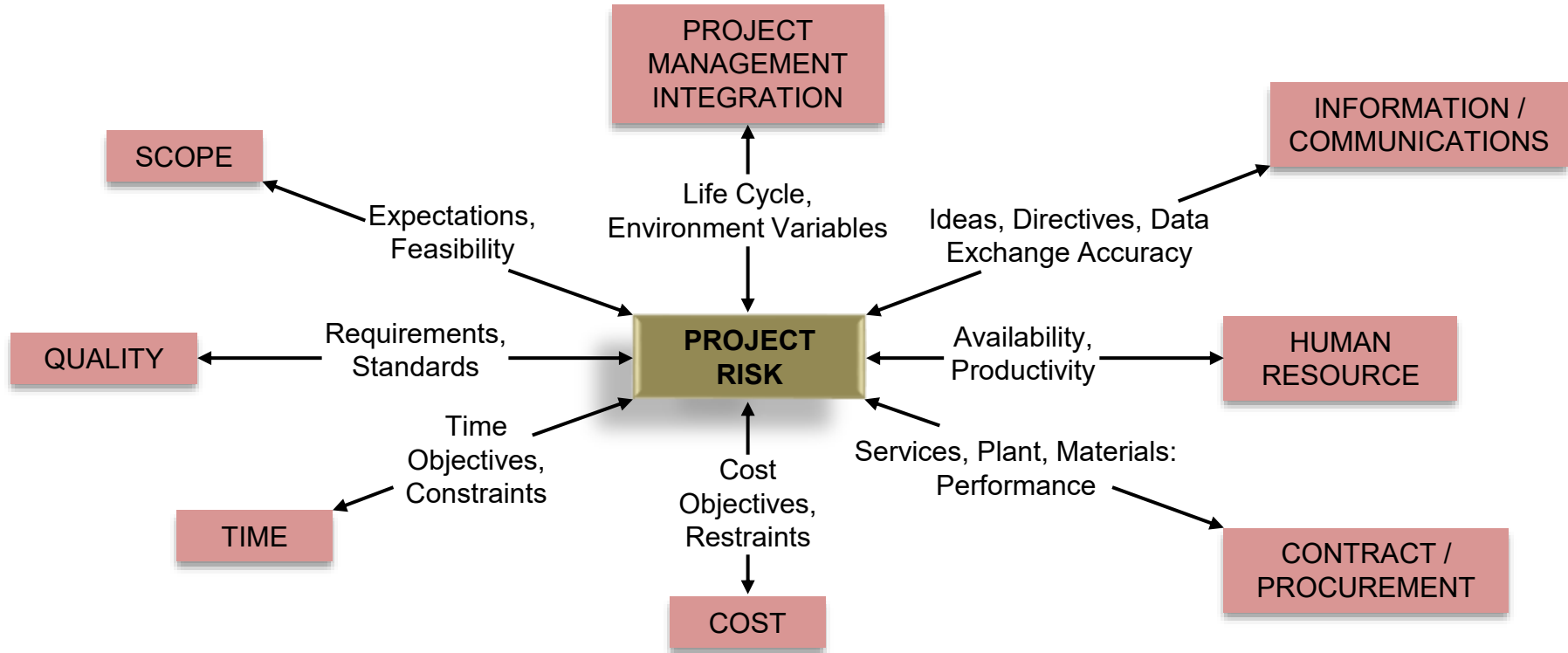


What is Risk Management?

- Risk Management is a process of identifying, analysing, prioritising and responding to risk events.
- Integration of risk management activities into your other project management functions.
- Developing responses to risk to meet your project objectives.
- Project risk management is PROACTIVE.



Integrating Risk with other Project Management Functions



Results from developing Risk Management Plan

- You have a written plan.
- You know what actions you have to do.
- You know who is responsible for what.
- You can track your work.
- You can learn from your risk activities and help others with their risk.

Risks vs Issues

- Many projects use risk and issue logs. Sometimes the management of issues and risks can become confusing.
- The PMBOK definition of an Issue:
 - **“A point or matter in question or in dispute, or a point or matter that is not settled and is under discussion or over which there are opposing views or disagreements.”**
- If you have the freedom to define these items and their logs, and the subsequent management of risks and issues; then great. Handle risks and issues as you desire. My suggestion is to follow the PMBOK guidelines as closely as possible .
- If you are dictated by the company, organisation or management team to handle risks and issues in a particular manner, then follow these guidelines. Document – in your Project Management Plan, Risk Management Plan and/or Issue Management Plan – how you will handle risks and issues.

Risk Identification



Risk in Corporate Business



Risk in corporate business is typically divided into two basic types:

- **Business Risk:** chances of profit or loss associated with a business endeavour.
 - Businesses employs qualified workers to increase profit and reduce chances of loss.
- **Pure or Insurable Risk:** divided into four categories:
 - **Direct Property:** destruction of property by fire ... etc.
 - **Indirect Property:** extra expenses associated with rental property or loss due to a business interruption.
 - **Liability:** chance of a lawsuit of bodily injury, damages ... etc.
 - **Personnel:** injuries to workers (workers' compensation).

Risk in Project Management

- Usually not enough attention is paid to risk on projects.
- All risks are not independent, and frequently the greatest risk on a project comes from a series of related/integrated events.
- Ultimate responsibility of risk management resides with the project sponsor.
- As the project manager (PM) is representing the sponsor, risk management becomes a large responsibility for PM.



Risk Identification Process



- Risk identification is never done.
- Risk identification is performed throughout the life of the project.
- The process for identifying risk:
 - Understand the project.
 - Identify the risk event.
 - Document the results and take appropriate actions.

What do you need to identify risk?

- **Product description**
- **Planning documents**
 - Project scope statement.
 - Cost management plan.
 - Schedule management plan.
 - Communications management plan.
 - Enterprise environmental factors.
 - Stakeholder register.
 - Quality management plan.
 - Organisational process assets.
- **Historical Information**
 - Previous project data.
 - Expert knowledge.



Group Exercise 1



- Consider a organising a music festival:

Identify 15 potential risks that may either positively or negatively impact on the successful delivery of that project.

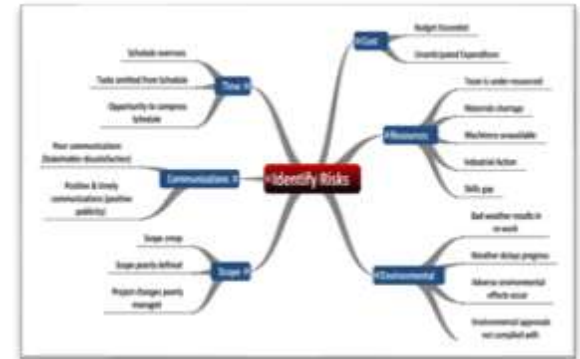
Risk Identification Meeting



- **Conduct a risk identification meeting**
 - Gather all relevant data.
 - Schedule a risk management meeting with your core team members.
 - Use a structured approach: Brainstorming, Nominal Group technique, Delphi technique, Mind Mapping, Project Lessons Learned ... etc.
 - Focus on identifying risk only.
- **Schedule risk identification meetings in your project plan**
 - After certain milestones: requirements complete, design complete ... etc.
- **Event driven**
 - A risk event happens and becomes part of the risk register.

Mind Mapping

- Begin with a category of risk in the center represented by, for example, a circle.
- Major risks for that category are represented by lines connecting with the circle.
- For each major risk, identify smaller risks that are part of that risk.
- Do not judge or evaluate at this time.
- Continue until no more risks can be identified.



SWOT Analysis



S

STRENGTHS

- Business-friendly; has high risk appetite and a record of entrepreneurship
- Prosperity of Gujarati NRIs and investors

O

OPPORTUNITIES

- With GIFT City and possible UNESCO World Heritage City status, poised to be an important Smart City if it delivers on implementation

W

WEAKNESSES

- Prone to floods and earthquakes
- Communally sensitive, although it has overcome its history of riots

T

THREATS

- Sensitive to global financial volatility
- Shares a border with Pakistan: the 1965 war impacted Ahmedabad

Delphi Technique

- Identify your risks in a risk register or a risk log:



Functional Area	Identify the functional business areas potentially impacted by the risk
Risk Category	Cost, External, Schedule, Technical, Resources, Operational
Risk Description	Description of the risk and the impact of it
Date Identified	Date the risk was identified
Raised By	Who identified the risk

Key Questions!

- **What are the right risks to manage?**
 - Analysing risks for probability and impact.
 - Developing a risk profile for your project.
 - Prioritising your risks.
- **When to quantify risks?**
 - Whenever a new risk is created.
 - An existing risk changes.
 - Influential factors change.
 - New information surfaces.
 - A change is proposed by the sponsor.
 - Market conditions change.
 - Significant personnel leave the project.



Quantitative Analysis vs. Qualitative Analysis

■ Quantitative Analysis

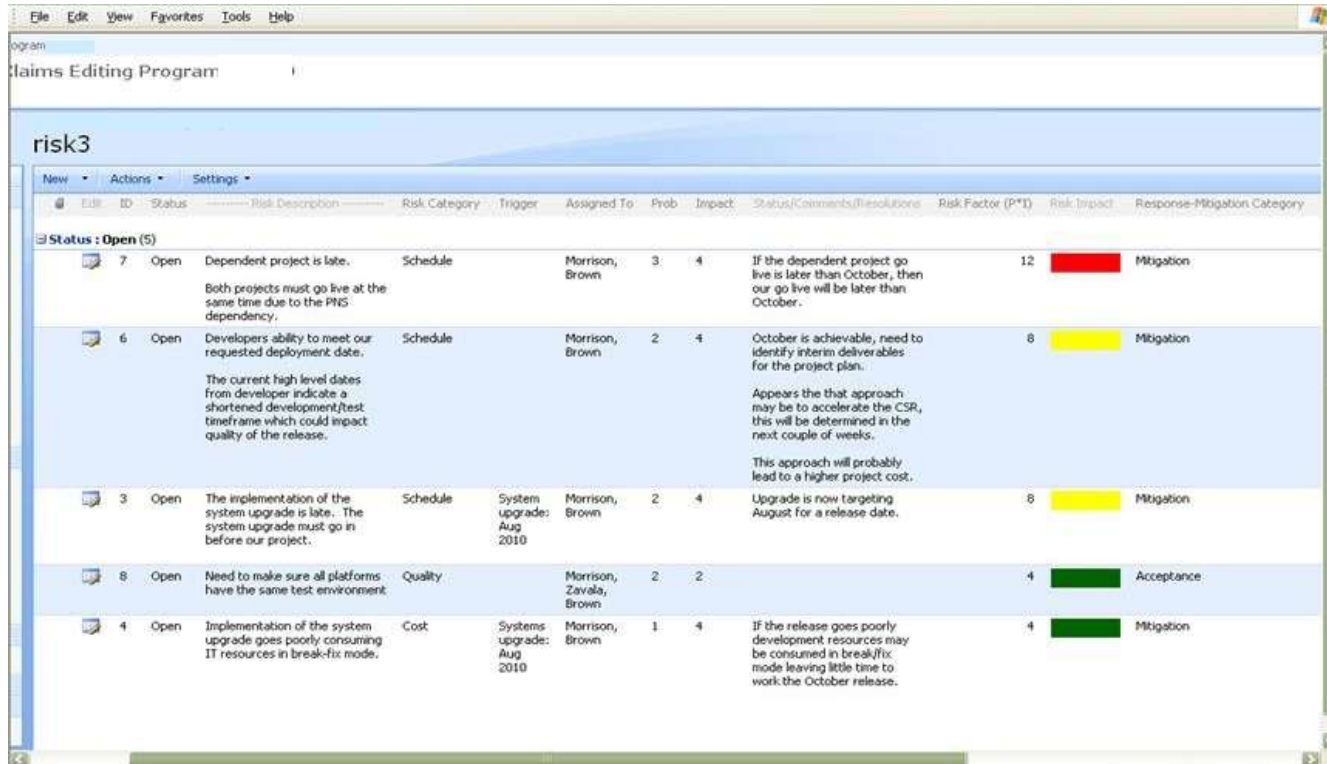
- Relies on a numeric value.
- Uses objective data.
- Requires understanding of probability theory.
- Removes some uncertainty.
- Should be based on historical data.
- Some examples are: sensitivity analysis, expected monetary analysis, and modeling and simulation.

■ Qualitative Analysis

- Uses subjective values: Green, Amber, Red.
- Requires common understanding of ordinal ranking system.
- May be less precise than quantitative analysis.
- Should be defined in terms of the parameters of the project.



Example of Risk Quantification (software)



ID	Status	Risk Description	Risk Category	Trigger	Assigned To	Prob	Impact	Status/Comments/Resolutions	Risk Factor (P*I)	Risk Impact	Response-Mitigation Category
7	Open	Dependent project is late. Both projects must go live at the same time due to the PNS dependency.	Schedule		Morrison, Brown	3	4	If the dependent project go live is later than October, then our go live will be later than October.	12	High	Mitigation
6	Open	Developers ability to meet our requested deployment date. The current high level dates from developer indicate a shortened development/test timeframe which could impact quality of the release.	Schedule		Morrison, Brown	2	4	October is achievable, need to identify interim deliverables for the project plan. Appears the that approach may be to accelerate the CSR, this will be determined in the next couple of weeks. This approach will probably lead to a higher project cost.	8	Medium	Mitigation
3	Open	The implementation of the system upgrade is late. The system upgrade must go in before our project.	Schedule	System upgrade: Aug 2010	Morrison, Brown	2	4	Upgrade is now targeting August for a release date.	8	Medium	Mitigation
8	Open	Need to make sure all platforms have the same test environment	Quality		Morrison, Zavala, Brown	2	2		4	Low	Acceptance
4	Open	Implementation of the system upgrade goes poorly consuming IT resources in break-fix mode.	Cost	Systems upgrade: Aug 2010	Morrison, Brown	1	4	If the release goes poorly development resources may be consumed in break/fix mode leaving little time to work the October release.	4	Low	Mitigation



Probability

- Can be done in a basic approach by developing a simple estimate of the probability of an event that will be late in delivery.
 - Ed says it is 50% likely this task will be late.
 - Probability of Event 1 x Probability of Event 2 = Probability.
- Can be done in a more complex manner by using weighted averages.
 - Joe says 35% chance of being late.
 - Mary says 40% chance of being late.
 - Ed says 50% chance of being late.
 - Joe gets twice as much credit, because he knows more about the situation.
 - The probability is: $((2 \times 0.35) + (0.4) + (0.5)) / 4 = 0.4 = 40\%$
- Quantifying risk probability can become quite complex; there are many resources to assist you with more detailed approaches (books, internet research, multi-day training, consultants ... etc.)



Assessing Impact (cont.)

- **Quality**

- Ask yourself the question “what if the project fails to perform as expected during its operational life?”.
- Of all the project objectives, conforming to quality objectives is the one most remembered.
- Therefore, this is one of the most important dimensions impacting your project.
- You can use financial analysis to identify risk for poor quality by quantifying long term activities that will impact the product lifecycle for your analysis.



Likelihood vs. Impact Matrix

Likelihood

- High: it is almost certain that the risk will happen (80-100% probability)
- Moderate: It is possible that the risk will occur (40-80% probability)
- Low: it is unlikely that the risk will occur except for in exceptional circumstances (0-40% probability)

Impact

- High: If this risk occurs it will have significant hindrance to the delivery of the programme
- Moderate: If this risk occurs it will cause some difficulty in the delivery of the programme
- Low: If this risk occurs it will have minimal impact on the delivery of the programme



	Impact →				
	Negligible	Minor	Moderate	Significant	Severe
Likelihood ↑	Very Likely	Low	Moderate	High	High
	Likely	Low	Moderate	High	High
	Possible	Low	Moderate	Moderate	High
	Unlikely	Low	Moderate	Moderate	Moderate
	Very Unlikely	Low	Low	Moderate	Moderate

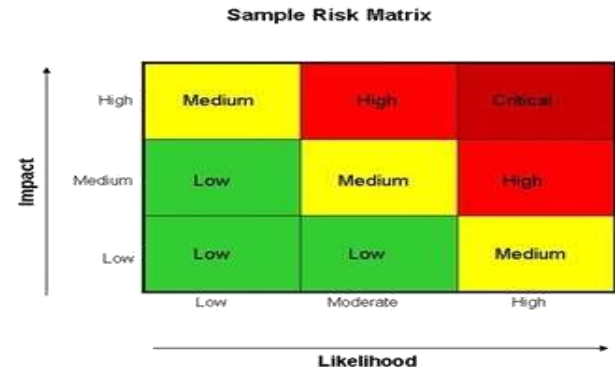
Issues at manufactures – cap on hand soap replaced for 6 weeks until new stock arrives

Heavy rain in Cornwall may cause flooding in some stores

Exercise 2: Individual

- Assess the probability and impact of the five risks identified in exercise 1.
- Briefly describe each risk, and then rate each one as high, medium or low in terms of probability and impact.
- Plot the results on a probability/impact matrix.

Duration: 10 minutes + 5 minutes for feedback.



Risk Response



Introduction

- The primary goal of the 'Plan' step is to prepare specific management responses to the threats and opportunities identified; ideally to remove/reduce the threats and to maximise the opportunities.



Risk Response consists of...

- Defining steps for responses to opportunities and threats.
- Assigning responsibility.
- Developing responses for **negative** risks or Threats:
 - **Avoid**: changing the project management plan to eliminate the risk. Could involve changing the objective, modifying the schedule or reduction in scope.
 - **Mitigate**: a reduction in the probability or impact to the project. Taking early action to reduce the probability, adopting less complex processes or conducting more tests.
 - **Transfer**: shifting the risk to a third party for the management of the risk. Does not eliminate the risk; could involve insurance, warranties, bonds ... etc.
 - **Insurance**: purchase insurance to reduce/eliminate risk – an athlete may purchase insurance against injury to guarantee their income.
 - **Accept**: it is possible that the risk cannot be eliminated or managed. Can be active or passive in approach – a contingency reserve in time, money or resources.



Risk Response consists of... (cont.)

- Developing responses for **positive** risks or opportunities
 - **Exploit** the situation. We will do whatever we can to make sure the event does happen, so we can enjoy the rewards of the event.
 - **Enhance** the probability and positive impacts of the event.
 - **Share** the ownership with a third party who can better enhance the situation.
 - **Accept** the opportunity; take the advantages provided by the event, but do not actively pursue the event.



Risk Response Tips

- Approach response development from a project wide perspective.
- Consider related risks.
- Stay within your project scope on your responses.
- Consider the following for contingency planning:
 - The management of a contingency budget.
 - The development of schedule alternatives and work-arounds.
 - Complete emergency responses to deal with major areas of risk.
 - An assessment of project shut-down liabilities.



Risk Budget

- Sum of money set aside.
- Contained within project budget.
- Funds specific responses to threats/opportunities.
- Financial approach required e.g. expected monetary value.
- Controlled (as defined in Risk Management Strategy).
- Provision for known/unknown risks?



Risk – Roles and Responsibilities

Role	Responsibilities
Corporate or programme management	Provide the corporate risk management policy and risk management process guide (or similar documents)
Executive	Be accountable for all aspects of risk management and, in particular, ensure a project Risk Management Strategy exists. Ensure that risks associated with the Business Case are identified, assessed and controlled. Escalate risks to corporate or programme management as necessary.
Senior User	Ensure that risks to the users are identified, assessed and controlled
Senior Supplier	Ensure that risks relating to the supplier aspects are identified, assessed and controlled.
Project Manager	Create the Risk Management Strategy Create and maintain the Risk Register Ensure that project risks are being identified, assessed and controlled throughout the project lifecycle.
Team Manager	Participate in the identification, assessment and control of risks.
Project Assurance	Review risk management practices to ensure that they are performed in line with the project's Risk Management Strategy.
Project Support	Assist the Project Manager to maintain the project's Risk Register.

Risk Analysis and Responses

- Which risks to act on/avoid?
“High” impact AND “High” probability.
- Which risks to monitor closely/mitigate or reduce?
“High” impact OR “High” probability.
- Which risks to review/accept?
“Medium” impact AND/OR “Medium” probability.
- Which risks to ignore/accept?
“Low” impact AND “Low” probability.



Risk Analysis and Responses (cont.)

Risk Assessment Matrix (3x3)				
IMPACT (consequence/severity)	High (We couldn't function or our mandate would have to change)	(3) Considerable management and monitoring required	(2) Manage and monitor risks (inform senior management)	(3) Extensive management (extensive senior management involvement)
	Medium (We could still function)	(3) Risk may be worth accepting with monitoring	(2) Management effort worthwhile, mitigate and monitor risks	(1) Must manage and monitor risk (inform senior management)
	Low (Normal)	(1) Accept risks	(2) Accept, but monitor risks	(1) Manage, mitigate and monitor risks
		Low (Normal or Unlikely)	Medium (Likely)	High (Very likely)
LIKELIHOOD (probability/frequency)				



Communicate

- **Communication is a step that is carried out continually**
 - The 'communicate' step should ensure that information related to the threats and opportunities faced by the project is communicated both within the project and externally to stakeholders.
- **Risks are communicated as part of the following management products:**
 - Checkpoint reports, bulletins, notice boards, dashboards, briefings ... etc.
 - Highlight reports.
 - End stage reports.
 - End project reports.
 - Lessons reports.
- **Communication aspects:**
 - Identification of risks.
 - Risk can change status.



Group Exercise 3

- Work with your probability impact matrix developed in exercise 2.
- Discuss one possible response to each one of the risks identified.
- Also, prepare an entry for one of the risks for the Risk Log.



Duration: 15 minutes + 5 minutes for feedback.

Risk Control



Implement/Control

- The primary goal of the 'implement' step is to ensure that the planned risk responses are actioned, their effectiveness is monitored, and corrective action is taken where responses do not match expectations.
- Important roles in this respect are:
 - **Risk Owner**: a named individual who is responsible for the management, monitoring and control of all aspects of a particular risk assigned to them; including the implementation of the selected responses to address the threats or to maximise the opportunities
 - **Risk Actionee**: an individual assigned to carry out a risk response action or actions to respond to a particular risk or set of risks. They support and take direction from the risk owner.

Risk Register/Log

- The purpose of the Risk Log is to capture and maintain information on all of the identified threats and opportunities relating to the project.
- Each risk on the Risk Log is allocated a unique identifier as well as details such as:
 - Who raised the risk.
 - When it was raised.
 - The category of risk.
 - The description of the risk (cause, risk event, effect).
 - Probability, impact and expected value.
 - Proximity.
 - Risk response category.
 - Risk response actions.
 - Risk status.
 - Risk owner.
 - Risk actionee.

Risk Register/Log

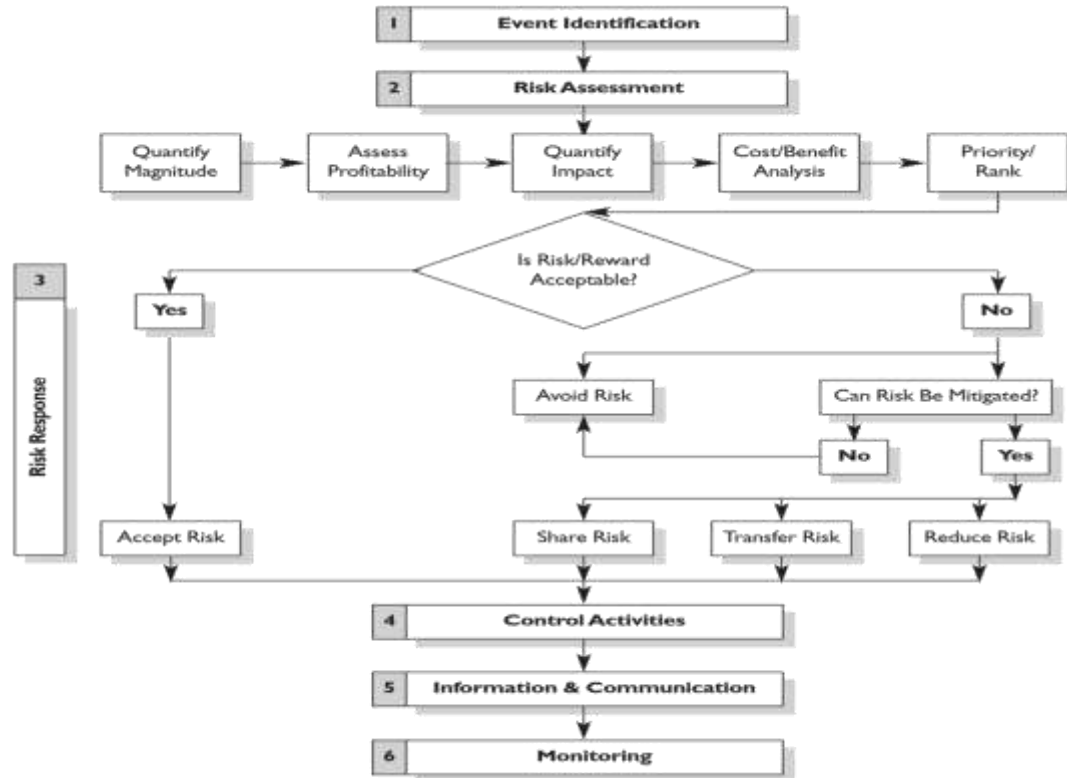
RISK	RISK DESCRIPTION	LIKELIHOOD OF THE RISK OCCURRING	IMPACT IF THE RISK OCCURS	SEVERITY RATING BASED ON IMPACT AND LIKELIHOOD	OWNER	MITIGATING ACTION	CONTINGENT ACTION
1	Quantitative survey does not reach proposed number of participants	MEDIUM	MEDIUM	MEDIUM	PROJECT LEAD	PROMOTE SURVEY USING A NUMBER OF SOCIAL MEDIA PLATFORMS TO INCREASE ITS REACH	IMPLEMENT AN ALTERNATIVE METHOD OF DATA COLLECTION
2	Difficulty in enrolling of collaborators showing little interest in project	LOW	HIGH	HIGH	PROJECT LEAD	CONDUCTED RESEARCH ON COLLABORATION OPPORTUNITIES PRIOR TO APPLICATION SUBMISSION	CONNECT WITH COLLABORATORS ON A GLOBAL SCALE
3	Lack of customer interest in final product	MEDIUM	LOW	MEDIUM	PROJECT LEAD	ENSURED PROJECT CAN SATISFY A RANGE OF OUTCOMES INCLUDING FURTHER ACADEMIC RESEARCH, FIBRE SALES, GOVERNMENT INTEREST IN BIO ECONOMY OPPORTUNITIES	RELATE FURTHER RESEARCH OPPORTUNITIES FOR THE PURPOSE OF THE RESEARCH DEVELOPMENT, SHARE INFORMATION WITH UKISHEN RETAILERS GLOBALLY
4	Shortage of suitable local materials	LOW	MEDIUM	LOW	PROJECT LEAD	COLLABORATED WITH THIRD PARTY COMPANIES FOCUSED ON PROMOTING THE LOCAL CIRCULAR ECONOMY AND EXPLORING OPPORTUNITIES IN LINE WITH THIS	EXTEND TO NATIONWIDE INITIALLY WITH A LONGER TERM GOAL TO USE LOCAL RESOURCES
5	Minimal access to material testing laboratories	LOW	HIGH	MEDIUM	PROJECT LEAD	EXTENSIVE RESEARCH ENLIGHTENED AVAILABLE LABS IN THE UK FOR FIBRE TESTING, LINKS WITH LOCAL FIBRE SPECIALISTS	EXTEND TO NATIONWIDE INITIALLY WITH A LONGER TERM GOAL TO WORK WITH LOCAL LABS
6	Work packages exceed allocated times	MEDIUM	LOW	MEDIUM	PROJECT LEAD	ALL WORK PACKAGES HAVE BEEN ALLOCATED TWO THIRDS AN INITIAL PLANNING DATE AND A CONTINGENCY DATE	IMPLEMENTATION OF A SPENDING CAP UNTIL WORK HAS CAUGHT UP
7	Work packages are completed in shorter periods of time than allocated	LOW	LOW	LOW	PROJECT LEAD	CROSSOVERS BETWEEN TASKS HAVE BEEN CONSIDERED WITHIN THE PROJECT PLAN	TEMPORARY PROGRESS HOLIDAY WILL NOT BE AN ISSUE
8	Repeated fibre testing without results	MEDIUM	HIGH	HIGH	PROJECT LEAD	EXISTING SUSTAINABLE MATERIALS CAN BE TESTED TO ASSESS IMPROVEMENTS RATHER THAN DEVELOPING A COMPLETELY NEW FIBRE	CONDUCT RESEARCH TO ASSESS EXISTING MATERIAL'S SUSTAINABILITY AND ACCESSIBILITY
9	Failure to complete consumer centric product designs	LOW	LOW	LOW	PROJECT LEAD	EXISTING UNDERSTANDINGS AND RESEARCH OF CONSUMER DESIGN NEEDS AND ISSUES	INCREASE THE SCOPE OF THE RESEARCH
10	Project Overspend	MEDIUM	MEDIUM	MEDIUM	PROJECT LEAD	PERFORMED VIABILITY TEST	APPLY FOR FURTHER CREDIT OR ALLOCATE PROFITS FROM EXISTING BUSINESS

Risk Register Example

Project Name:	XX	No. of Risks	Total Approx £ impact of Risks at level
Project No:	XX	1	£4,800.00
Project Manager:	XX	1	£3,960.00
Project Board:	XX	2	£1,824.00

[illegible]

Risk Management Process



Risk Log Example

Risk ID	Sequential number assigned
Functional Area	Identify the functional business areas potentially impacted by the risk
Risk Category	Cost; External; Schedule; Technical; Resources; Operational
Risk Description	Description of the risk and the impact of it
Date Identified	Date the risk was identified
Raised By	Who identified the risk
Date Assigned	Date the risk was assigned
Assigned To	Who the risk was assigned to
Probability	1, 2, 3, 4
Potential Impact	1, 2, 3, 4
Risk Factor (P*I)	Probability * Impact
Positive or Negative Impact	Will the potential impact of the risk have a Positive, Negative, Both or Unknown impact if realized?
Response Category	Acceptance; Mitigation; Transfer; Avoidance
Status/Comments	Status of risk and update/comments about it
Trigger	Preliminary event that will indicate the risk is about to take place
Proposed/Actual Resolution	Risk Response plan
Contingency Plan	Alternate Plan if Risk Response fails



Conclusion

- Risk Management requires:
 - Planning.
 - Structure.
 - Analysis.
 - Creativity.
 - Constant attention.
 - Flexibility.
 - Communications, communications, communications!!!

Conclusion (cont.)

- The PMBOK has a lot of great reference materials to assist you with your Risk Management planning activities.
- Whatever you do, just do SOMETHING to address risk on your projects – and do it in a structured manner.
- If your company/organisation does not have templates or a process in place, develop your own tools.
 - Use the PMBOK for ideas on the approach and build a risk register in Excel.
 - A tab in Excel identifying your “plan” and another tab with the Risk Register will work much better than nothing.

Conclusion (cont.)

- Program Risk Management
 - Project risk management principles can be applied to programs.
 - Use of an RBS will be helpful for programs.
 - Project risks will roll up to the program.
 - Program risks will take on a more global nature in addition to the project risks within the program.
 - External events, company profits, legal/compliance issues, political issues, company strategies ... etc.
 - Communication with your sponsors is even more critical on programs than on projects.

Any question?